Yerdaulet Zhumabay

Almaty, Kazakhstan

yerdaulet.zhumabay@gmail.com | yerdaulet-zhumabay | github.com/Yerdaulet-zh

Education

Almaty Management University **Aug 2018 – Jun 2022**
Bachelor of Business Analytics and Big Data *Almaty, Kazakhstan*

Kazakh-British Technical University **Aug 2022 – Jun 2024**
Master's Degree of Data Science *Almaty, Kazakhstan*

- **Conference on paper:** Title: "Attention on Attention as a Part of Decoder for Vehicle License Plate Recognition"
  The 2024 IEEE AITU: Digital Generation conference, held on April 3, 2024, Astana, Kazakhstan.
  The AROB-ISBC-SWARM 2024 symposium, which took place from January 24-26, 2024, at the B-Con Plaza in Beppu, Japan.

Experience

Jun 2021 – Aug 2021
Intern Data Miner *Almaty, Kazakhstan*

- Contributed to the development of advanced personal recommendation systems for Kaspi Store by analyzing user data and utilizing machine learning ALS, BPR algorithms.
- Trained both explicit and implicit-based algorithms on large-scale personal data from the user database to enhance user experience and recommendation accuracy.
- Technologies utilized: Python, SQL, Jupyter notebook and Linux

May 2024 – Present
AI and MLOPS Engineer *Almaty, Kazakhstan*

- Led the development and deployment of classical Arabic speech transcription AI model, improving transcription accuracy by fine-tuning whisper v3 in low data environment.
- Optimized transformer-based whisper v3 architecture source code, applied quantization techniques using TensorRT, achieving solid 3x performance of inference.
- Implemented a custom whisper v3 encoder - based architecture trained by CTC loss with MLFlow and DVC, reaching on the single Nvidia T4 GPU with 8vCPUs 700 RPS* for 5 seconds lasting audio.
- Managed a team of audio annotators, configured Labelbox, later open-source Label Studio deployment on AWS EC2 for annotators ensuring high-quality annotations of training data.
- Utilized Nvidia Triton Inference Server and FastAPI to serve AI models.
- Managed AWS SageMaker for AI API deployment and serving, API Gateway with token based authorization by AWS Lambda, ensuring API scalability and performance.
- Managed Docker Swarm for micro-service clustering, configured Nginx (Reverse-proxy), Certbot, PostgreSQL, Prometheus, Loki, and Grafana by reducing cost of AI API serving in production more than 30%* compared to AWS SageMaker real-time inference.
- Terraform to prepare and manage AWS Services for AI micro-service.
- Ansible for configuration and setup of AWS EC2 (Linux, Ubuntu) instances.

Technical Stack

Cloud Infrastructure: AWS (VPC, IPAM, EC2, S3, KMS, Lambda, IAM, Control Tower, Identity Center, Route53, Service Quotas), GCP (VPC, Artifact Registry, Compute Instances)

Containerization & Orchestration: Kubernetes (AWS EKS, GCP GKE), Karpenter, KEDA, Cluster Autoscaler, Cert-Manager, AWS Load Balancer Controller, Nginx Gateway Fabric, Redis Cluster, Docker, Docker Compose, Docker Swarm

Infrastructure as Code & GitOps: Terraform, Terragrunt, Ansible, ArgoCD, Git, GitLab CI, Jenkins

Observability & Telemetry: Grafana, Grafana Alloy, Prometheus (kube-prometheus-stack), Loki, Tempo, SonarQube

MLOps & AI Infrastructure: Triton Inference Server, TensorRT, ONNX, MLFlow, DVC, AWS SageMaker

Programming & Frameworks: Python (FastAPI, NumPy, Pandas, Scikit-Learn, PyTorch, ONNX, TensorRT, Torchaudio, Albumentations, Pillow, OpenCV), Golang (net/http, GORM)

Databases & Backend Tools: PostgreSQL, Redis, SQLAlchemy, Alembic, Atlas CLI, Nginx

Deep Learning Core (Concepts): CNNs, Vision Transformers (ViT), YOLO, Transformers (BERT, RoBERTa, WhisperV3), Specific Loss Functions (CTC, ArcFace, CosFace)